PRACTICAL APPLICATION:

5. Evidence Analysis:

Evidence analysis means examining the collected digital evidence to find useful facts about what happened, when it happened, how it happened, and who was involved.

Evidence Analysis:

Analyzing network connections from a windows vm using velociraptor to identify suspicious connection,

after installing velociraptor from official site then goto downloads and run command prompt and run

velociraptor.exe gui

then in browser

http://127.0.0.1:8889

velociraptor opens,

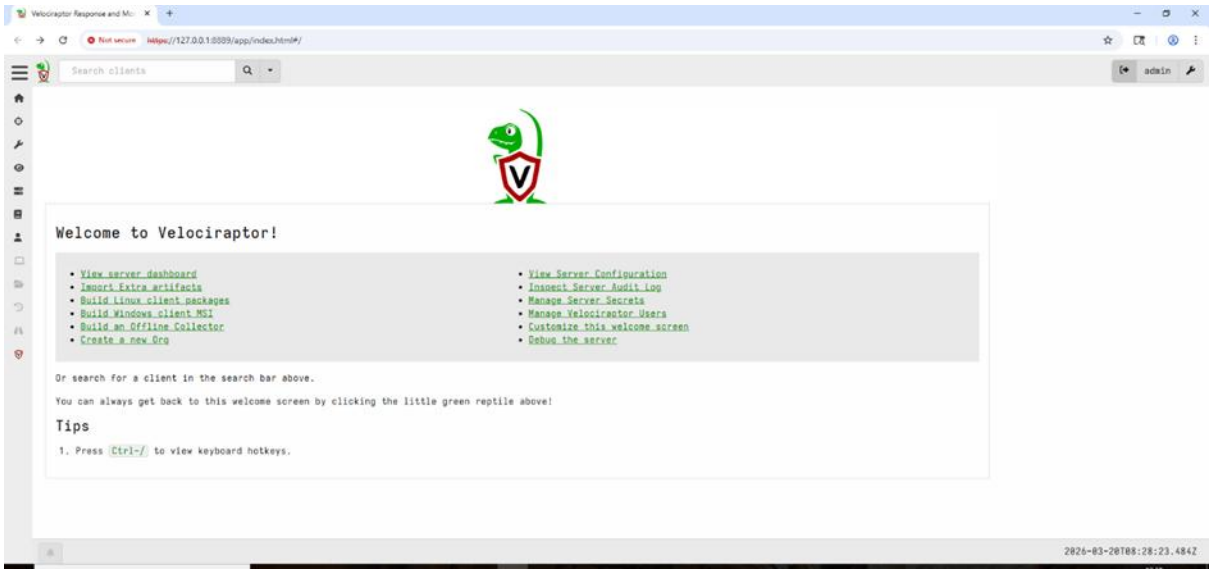

Figure 1: velociraptor accessing through browser

- Goto notebook then click on new notebook
- Then give name as netstat investigation-5
- Then click on launch
- After that goto add cell and insert cellby clcicking add cell then select VQL
- After that edit cell there by pasting below query

  SELECT * FROM netstat()
- Then clich save and run

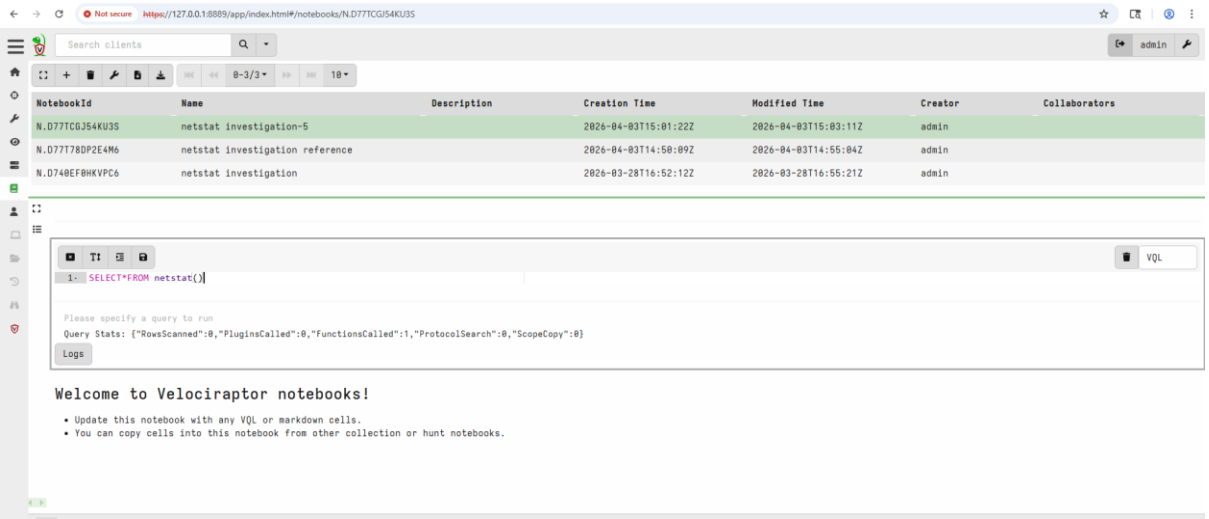

Figure 2: running query to analyze network connections

After running query it gaves results as below

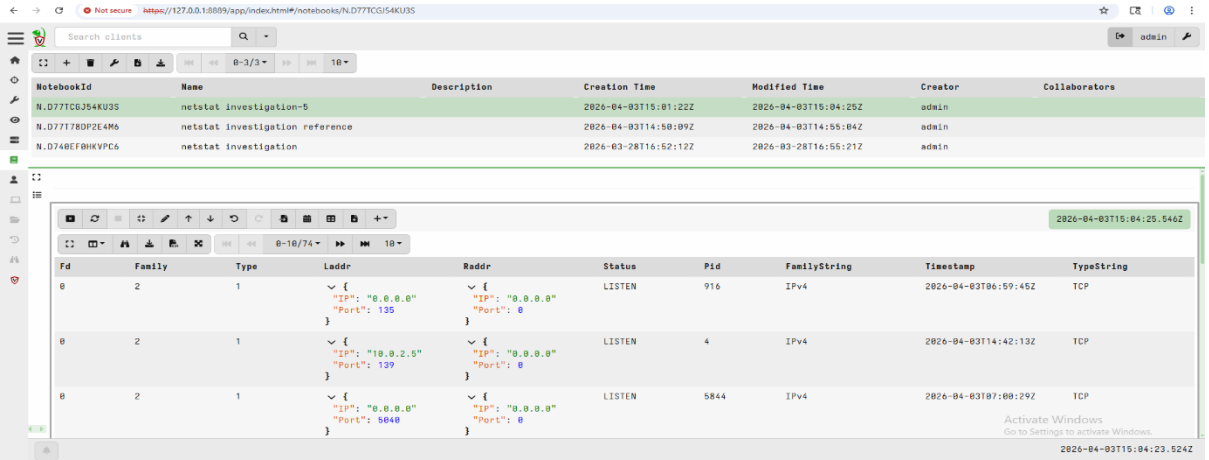

Figure 3

Figure 4

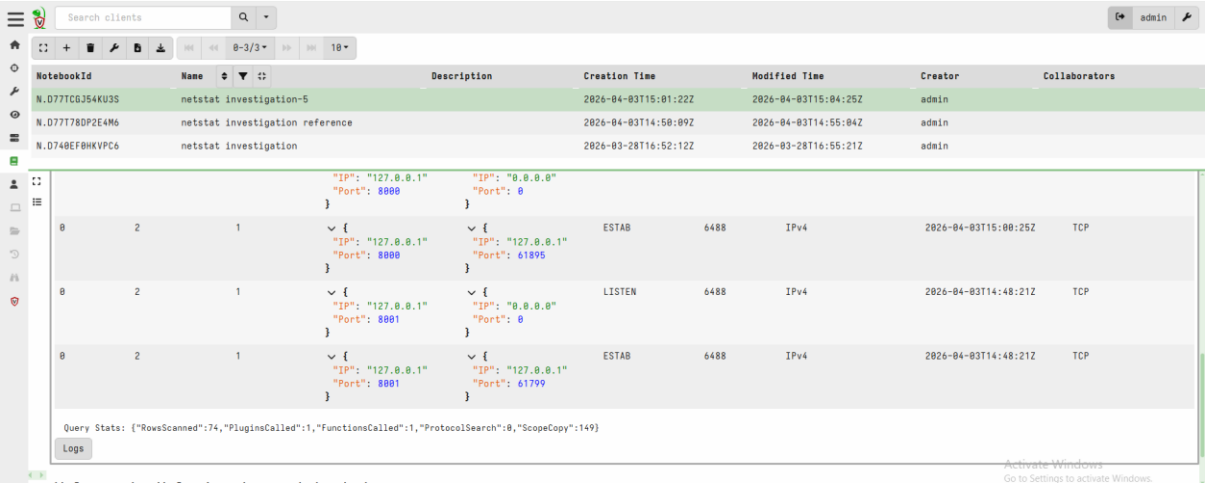

Figure 5

Figure 3,4,5, shows that The Velociraptor netstat analysis was reviewed to identify active and listening network connections on the Windows virtual machine. Most observed connections appeared to be local system or service-related activity.

Multiple listening ports were identified such as port 135 on 0.0.0.0, port 139 on 10.0.2.5, port 5040 on 0.0.0.0, port 8000 on 127.0.0.1, port 8001 on 127.0.0.1 . Several established connections were found on 127.0.0.1.

Velociraptor netstat output showed several listening and established connections. Most activity was related to local host communication 127.0.0.1 and common Windows service ports such as 135, 139, and 5040. No clear suspicious external connection was identified during the analysis.

After the output exported results to csv file ,
Click export to csv -→ click on download

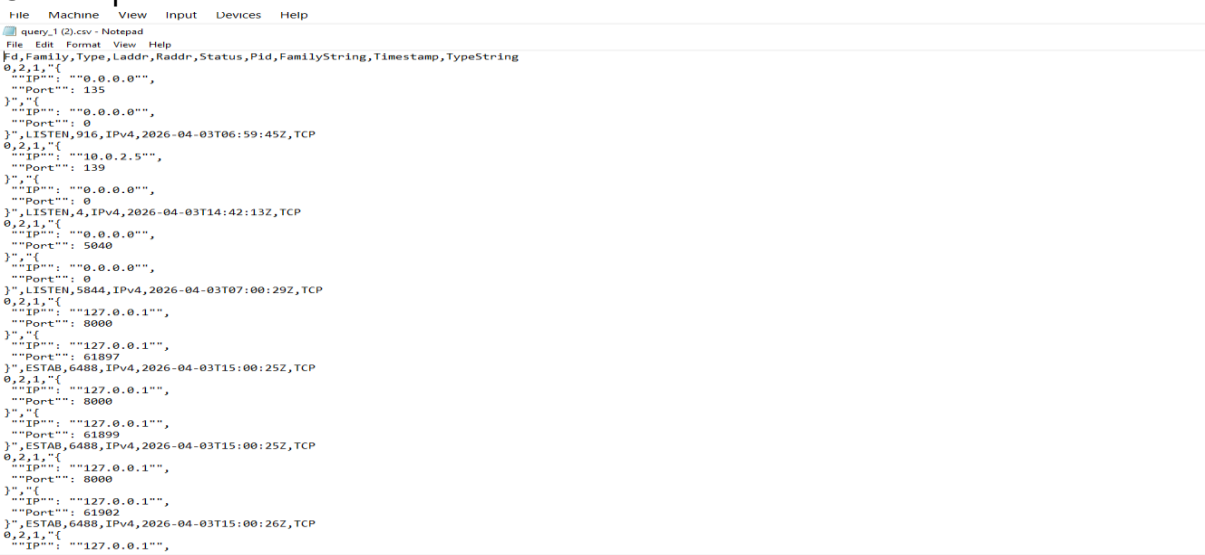

Figure 6: csv file of output

Create hash for downloaded csv file

Open powershell as administrator and create hash of downloaded csv file using ,

GET-Files -Path “C:\Users\windows\Downloads\query_1 (2).csv”

This will create the hash .

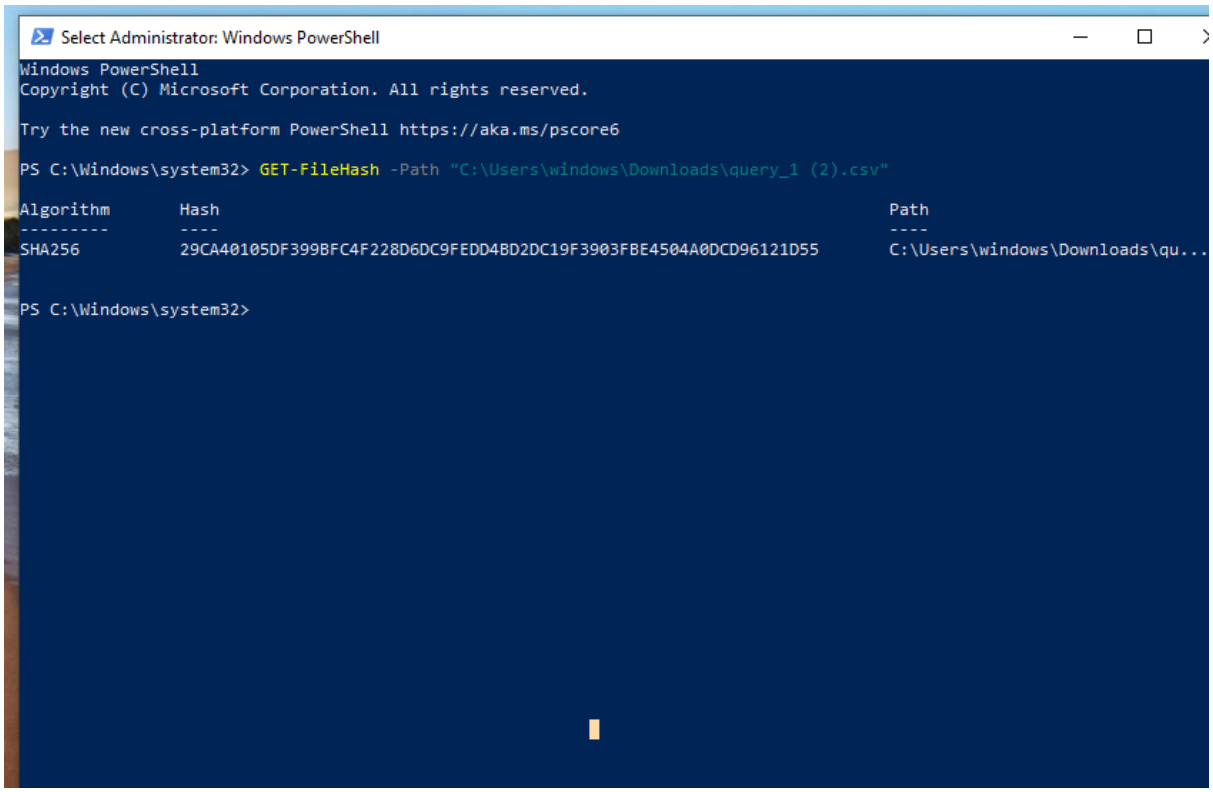

Figure 7: created the hash of downloaded output file

Chain-of-Custody:

Documenting evidence collected

Item	description	Collected by	date	Hash value
Network connection log	Analyzing Windows network connections	SOC analyst	03-04-2026	29CA40105DF399BFC4F228D6DC9FEDD4BD2DC19F3903FBE4504A0DCD96121D55

The network connection log was collected from the Windows system to analyze active and listening network activity. The evidence was collected by the SOC Analyst on 03-04-2026 and documented to ensure integrity and accountability throughout the investigation process. A SHA-256 hash value was generated and recorded to verify that the evidence remained unchanged after collection.

Summary:

Collected and analyzed Windows network connection evidence using Velociraptor to identify any suspicious or unauthorized communication. The review of the netstat output showed several active and listening ports, including common Windows service ports and localhost connections. Most observed activity appeared to be normal system or internal application communication, with no clear evidence of suspicious external network connections during the analysis.

To preserve integrity, the network connection log was properly documented as evidence and included in the chain of custody. A SHA-256 hash value was recorded to ensure the evidence remained unchanged after collection.